

Notes: Engelberg, Reed & Ringgenberg (JFE, 2012)
How are shorts informed? Short sellers, news, and information processing

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1 Big picture

- **Question.** Short sellers are known to be informed traders, but how do they become informed? Do they anticipate private information, manipulate or create rumors, exploit liquidity, or process public news better than other traders?
- **Core answer.** A large part of the short sellers' trading advantage comes from processing public news, not from systematically anticipating news before it is released.
- **Main empirical pattern.** The negative relation between short selling and future returns is much stronger on news days and strongest on days with negative news.
- **Economic magnitudes.** News days are only about 22% of sample days but account for more than 45% of the total profitability associated with short selling.
- **Trading strategy result.** A long-short strategy based only on short selling earns about 40% over the 2.5-year sample; conditioning on short selling and any news earns about 60%; conditioning on short selling and negative news earns about 180%.
- **Timing result.** Short sellers do not, on average, build abnormal positions before public news releases. Their relative short selling rises mainly on or after news days.
- **Who is informed?** Non-exempt short sales, which are closer to client trades, contain the information. Exempt short sales, which are closer to market-maker offsetting activity, do not explain the main results.
- **Alternative explanations.** The paper rejects two leading alternatives: that market makers are merely offsetting uninformed buyer demand, and that news days are simply cheaper liquidity days for short sellers.
- **Takeaway.** Public news is not automatically processed equally by all market participants. Short sellers appear to have a comparative advantage in interpreting public information quickly and profitably.

2 Incremental contribution of the paper

- **From whether shorts are informed to how they are informed.** Prior work showed that high short interest or short volume predicts low future returns. This paper asks the next question: what is the source of that information advantage?
- **Comprehensive news universe.** Earlier papers often focused on specific events such as earnings announcements, analyst downgrades, or fraud revelations. This paper links the full Dow Jones corporate news archive to daily short-sale data, allowing the authors to study all corporate news rather than a selected event type.
- **Timing around news.** The paper distinguishes anticipation from interpretation. If shorts know news before the market, short volume should rise before the news. If they process news better, profitability should rise on and after news. The data support the second channel.
- **Market-maker versus client distinction.** The exempt/non-exempt trade marking in the short-sale data allows the paper to separate likely market-maker offsetting trades from more information-motivated client short sales.
- **News content matters.** The paper shows that short sellers' predictability is strongest on negative news days, and then drills down by 39 news categories. This connects informed short selling to the economic content of information events.
- **Market quality tests.** The paper directly tests whether news days simply lower trading costs. The evidence points in the opposite direction: bid-ask spreads rise on news days, and rebate rates and Amihud illiquidity do not show broad improvements.
- **Broader conceptual contribution.** The paper provides evidence for the idea that information processing skill matters even when the raw information is public. Public disclosure does not fully equalize information unless all traders can interpret the disclosure equally quickly and accurately.

3 Data and measurement

3.1 Short-sale data

- The short-sale data come from the NYSE Regulation SHO/TAQ database.
- The sample period is January 3, 2005 through July 6, 2007.
- The sample covers NYSE-listed and traded securities, including U.S. common stocks, ADRs, REITs, and closed-end funds.
- The merged daily sample contains 1,888,868 firm-day observations and 3,167 securities.
- The main short-selling variable is

$$\text{ShortVolRatio}_{it} = \frac{\text{short volume}_{it}}{\text{total volume}_{it}}.$$

- The mean short-volume ratio is 0.196, so about one-fifth of trading volume is short selling in the sample.
- The database separates **exempt** trades from **non-exempt** trades. Exempt trades are used as a proxy for market-maker activity; non-exempt trades are used as a proxy for client short selling.

Interpretation: The exempt/non-exempt split is important because it lets the paper ask whether the results reflect informed client trades or market makers mechanically shorting to supply liquidity.

3.2 News data

- The news data come from the Dow Jones archive.
- Each story includes a release time, a firm identifier, stock symbols, and subject codes.
- Dow Jones subject codes are cleaned into 39 economically meaningful news categories after removing generic tags such as broad company-news indicators.
- If an article is released before the 4:00 PM market close, it is assigned to that trading day. If it is released after 4:00 PM, it is assigned to the next trading day.
- Multiple updates to the same story are treated as one unique news event, but the database also tracks the number of article updates.

Interpretation: The precise intraday timestamp makes the timing test meaningful. The paper can distinguish short sales before public news from short sales on or after news release.

3.3 News sentiment and news categories

- The paper uses two dictionary-based sentiment measures: Harvard-IV-4 negative words and the Loughran-McDonald negative word list.
- For the main timing and trading tests, the paper often signs news using the announcement-day return.
- A **negative news event** is a news day whose announcement-day return lies in the bottom quintile of all announcement-day returns.
- A **positive news event** is a news day whose announcement-day return lies in the top quintile.
- The paper also studies 39 Dow Jones news categories separately, including earnings, analyst comments, dividends, lawsuits, bond ratings, and new products.

Interpretation: Using announcement-day returns to sign news captures the market's surprise relative to expectations. A story with many negative words may not be bad news if investors expected something worse.

3.4 Returns and market quality

- The main outcome is the buy-and-hold return from day $t + 1$ to day $t + 20$ after the short-sale or news event.
- Returns are measured as raw returns, market-adjusted returns, and size- and momentum-adjusted returns.
- Market quality variables include equity loan rebate rates, Amihud illiquidity, and bid-ask spreads.
- Rebate rates measure the cost of borrowing shares. Lower rebates mean more expensive shorting.
- Amihud illiquidity and bid-ask spreads proxy for trading costs and liquidity around news events.

Interpretation: Market quality variables are used to test whether short sellers are simply more profitable on news days because news days are cheaper or more liquid trading days. The evidence does not support that explanation.

4 Models and empirical specifications

4.1 Timing of short selling around news events

The first family of tests asks whether short selling rises before news, on the news day, or after news:

$$\text{ShortVolRatio}_{i,t+k} = \alpha + \beta_1 R_{i,t-1} + \beta_2 R_{i,t-2} + \beta_3 \text{News}_{it} + FE_i + \varepsilon_{it},$$

where $k \in \{-2, -1, 0, +1, +2\}$ indexes event time relative to the news day.

Interpretation: If short sellers anticipate news, ShortVolRatio should rise before $t = 0$. If they process public news quickly after release, the strongest changes should appear on or after the news day.

4.2 Return predictability from short selling

The baseline cross-sectional return regression is:

$$\text{Ret}_{i,t+1,t+20} = \alpha + \beta_1 \text{ShortVolRatio}_{it} + \beta_2 \text{Size}_{i,t-1} + \beta_3 R_{i,t-1} + \beta_4 R_{i,t-2} + FE_i + \varepsilon_{it}.$$

The coefficient β_1 measures whether high short volume predicts lower future returns.

Interpretation: A negative β_1 means that stocks with more short selling today subsequently underperform, consistent with short sellers having information.

4.3 News interactions

The main specification interacts short selling with news indicators:

$$\text{Ret}_{i,t+1,t+20} = \alpha + \beta_1 \text{ShortVolRatio}_{it} + \sum_{n=1}^3 \gamma_n \text{NewsType}_{n,it} + \sum_{n=1}^3 \delta_n (\text{ShortVolRatio}_{it} \times \text{NewsType}_{n,it}) + \text{Controls} + \varepsilon_{it}.$$

The three news types are negative, neutral, and positive news events.

Interpretation: The interaction terms test whether short volume is especially informative on news days. The coefficient on $\text{ShortVolRatio} \times \text{NegativeNews}$ is the key test of whether short sellers process bad news particularly well.

4.4 Exempt versus non-exempt short sales

The paper repeats the return-predictability regressions separately for:

$$\text{ShortVolRatio}_{it}^{\text{nonexempt}} = \frac{\text{non-exempt short volume}_{it}}{\text{total volume}_{it}},$$
$$\text{ShortVolRatio}_{it}^{\text{exempt}} = \frac{\text{exempt short volume}_{it}}{\text{total volume}_{it}}.$$

Interpretation: If the results are driven by market makers offsetting uninformed buying pressure, exempt trades should explain the findings. If they are driven by informed short sellers, non-exempt trades should carry the signal. The paper finds the latter.

4.5 Portfolio return tests

The paper also forms daily long-short portfolios. A generic strategy sorts firms into quintiles based on yesterday's short-volume ratio and/or news status:

$$\text{PortfolioReturn}_t = \text{Return}(\text{LongLowShorting}_t) - \text{Return}(\text{ShortHighShorting}_t).$$

Interpretation: The portfolio exercise translates regression coefficients into economic magnitudes. It shows how much a trader could have earned by combining short-sale information with public-news information.

4.6 Market quality tests

To test the liquidity explanation, the paper examines event-time changes in:

$$\text{RebateRate}_t, \quad \text{Amihud}_t = 10^7 \frac{|r_t|}{\text{Volume}_t}, \quad \text{BidAskSpread}_t.$$

Interpretation: If the profitability of short selling on news days were due to lower costs, one should see cheaper borrowing, lower illiquidity, or narrower spreads. Instead, spreads rise on the news day and other market-quality measures do not improve.

5 Identification and empirical design

5.1 Identification question

The paper does not run a randomized experiment. Its empirical design is a set of event-time, cross-sectional, and portfolio tests designed to distinguish four mechanisms:

1. **Anticipation:** short sellers know or create news before it becomes public.
2. **Information processing:** short sellers interpret public news better than other traders.
3. **Uninformed counterparty:** market makers short because buyers make mistakes around news.
4. **Liquidity:** short selling is profitable on news days because trading costs are lower.

Interpretation: The identification strategy is to ask which mechanism makes the most precise predictions for timing, trader type, return predictability, and market quality.

5.2 Testing anticipation

The key timing prediction is:

H_1 : If short sellers anticipate news, $\text{ShortVolRatio}_{t-2}$ or $\text{ShortVolRatio}_{t-1}$ should rise before news.

The paper estimates event-time short-volume regressions and plots short volume around news events.

Interpretation: The evidence does not show a systematic pre-news increase in the short-volume ratio. Short sellers appear to trade on or after public news rather than before it.

5.3 Testing information processing

The main prediction is:

$$H_2 : \beta_{\text{ShortVolRatio} \times \text{News}} < 0,$$

especially for negative news. In words, high short selling should predict more negative future returns when public news has just been released.

Interpretation: This is the central empirical result. The negative relation between short selling and future returns is more than twice as large on news days and roughly quadruples on negative-news days.

5.4 Separating clients from market makers

The paper uses exempt versus non-exempt short-sale flags:

client/informed channel $\Rightarrow$ stronger results for non-exempt trades,

market-maker offsetting channel $\Rightarrow$ stronger results for exempt trades.

Interpretation: Non-exempt trades drive the findings. Exempt trades do not generate the same pattern, rejecting the idea that the main results are simply market-maker offsetting trades caused by uninformed buying pressure.

5.5 Testing the liquidity explanation

The liquidity hypothesis predicts that shorting costs or transaction costs improve around news:

$$H_4 : \Delta \text{RebateRate} > 0, \quad \Delta \text{Amihud} < 0, \quad \Delta \text{Spread} < 0.$$

Interpretation: The paper finds no broad improvement in rebate rates or Amihud illiquidity, and bid-ask spreads increase on news days. This makes a lower-cost-of-shorting explanation unlikely.

5.6 Controls and design choices

- Firm fixed effects control for time-invariant heterogeneity across stocks.
- Size and lagged returns control for known predictors of short selling and future returns.
- Newey-West standard errors address time-series dependence in daily cross-sectional regressions.
- To reduce news clustering, the paper removes stories in the same category that occur within 30 days of another story for the same firm.
- Category-specific regressions ask which kinds of news drive the result.

Interpretation: The paper's credibility comes from triangulation: timing tests, interaction tests, trade-type tests, liquidity tests, and category-specific tests point toward the same information-processing interpretation.

5.7 Main threats to interpretation

- **Reverse causality:** short selling could cause news coverage rather than respond to it. The paper's timing and future-return tests reduce but do not completely eliminate this concern.
- **Omitted variables:** news days may coincide with other unobserved signals. The use of firm fixed effects, controls, and category tests helps, but this is not a randomized design.
- **Classification error:** exempt and non-exempt tags imperfectly separate market makers from clients. The distinction is useful but not perfect.
- **Transaction costs:** portfolio profits are not net of all trading costs. The portfolio exercise measures gross economic signal strength rather than a fully implementable arbitrage.

Interpretation: The paper is strongest on mechanism: it shows that the short-selling signal becomes much stronger when public news is available and that the pattern is concentrated in likely informed client trades.

6 Tables and figures

6.1 Table 1: Dow Jones archive; Figure 1: Example time-series of short sales, news, and 20-day portfolio returns

Read:

- Table 1 shows the raw structure of the Dow Jones archive: story text, timestamps, stock symbols, and subject codes.
- The example illustrates why the archive is useful: it links specific firms, release times, and content categories.
- Figure 1 uses Williams-Sonoma as an example, plotting news days, short-volume ratio, and future 20-day adjusted returns.
- The example is not an estimate; it visually motivates the idea that short-sale intensity and news timing can be combined at daily frequency.

Detailed interpretation: The table makes clear that the authors can identify a public-information event precisely enough to compare short sales before, on, and after the release. The figure shows the empirical object at the firm-day level: some news days are followed by large changes in future returns, and short volume varies sharply around these events.

Economic message: The paper is built on the ability to merge public news with short-sale transactions at the daily firm level. These two visuals show the raw data structure and the kind of event-time variation used throughout the paper.

Table 1

Dow Jones archive.

The table provides an example of the information contained in the Dow Jones archive. Panel A displays the text of an example article, while Panel B contains the associated entry in the Dow Jones archive. The archive entry contains the date and time the article was released as well as the stock symbols of firms mentioned in the article and a series of subject codes that identify the content areas of the article. For example, the code RND indicates that the article pertained to research and development. Articles can have more than one time stamp, indicating that the article was updated following its initial release.

Panel A: Example news article

GlaxoSmithKline and EPIX Pharmaceuticals Enter Drug Discovery and Development Alliance	
DOW Jones Newswires	
EPIX Pharmaceuticals Inc. (EPIX) said it entered into a drug discovery and development pact with GlaxoSmithKline (GSK). As part of deal, EPIX will receive an upfront payment of \$35 million, which includes \$17.5 million from the sale of 3 million shares of its common stock. EPIX will also be eligible for up to \$1.2 billion for the achievement of certain milestones, and royalties on product sales. EPIX shares closed Monday unchanged at \$5.52 and Glaxo shares fell 9 cents to \$52.43. EPIX said it expects to end 2006 with more than \$100 million in cash and marketable securities. The company expects that its existing cash and marketable securities together with the expected revenue from the GlaxoSmithKline collaboration and other partnerships will be sufficient to fund operations through 2008.	
Panel B: Dow Jones archive values for the example article	
Story code=20061212003980	
Date=12/12/2006	
Time=06:12:00:29, 06:12:02:36, 06:12:16:34, 06:12:24:59	
Stock symbols=EPIX, GSK	
Subject codes=CNW, DJEN, DJGP, DJCS, DJCV, DJI, DJIN, DJIV, FCTV, WEI, RND, HDL	

(a) Table 1: Dow Jones archive. Original example of article text and machine-readable archive fields.

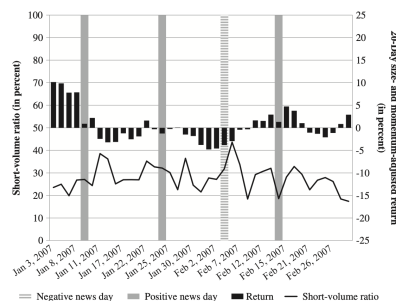


Fig. 1. Example time-series of short sales, news, and 20-day portfolio returns. The figure displays portfolio returns (right axis) and the short-volume ratio (left axis) each day for one example firm, Williams-Sonoma, from January 3, 2007 through February 28, 2007. Portfolio returns are the buy and hold (compounded) return percent over the subsequent 20 trading days and are calculated using size- and momentum-adjusted returns. Short-volume ratio is daily short volume/total volume. Negative (positive) news days are defined as days on which a news article about Williams-Sonoma is released and the firm's announcement day return is in the bottom (top) quintile of all returns that day, respectively.

(b) Figure 1: Example time-series of short sales, news, and 20-day portfolio returns. The figure shows how news timing, future returns, and the short-volume ratio can be aligned in event time.

6.2 Table 2: Summary statistics; Table 3: Regression analysis of short-volume ratio around news events

Read:

- Table 2 reports 1,888,868 daily observations and 3,167 securities.
- The mean short-volume ratio is 0.196; non-exempt trades dominate exempt trades.
- The largest news categories include high-yield issuers, acquisitions, analysts' comments, earnings, earnings projections, and personnel appointments.
- Table 3 shows that short sellers do not meaningfully increase relative shorting before public news events.
- On negative news days, the short-volume ratio at $t = 0$ is lower rather than higher, inconsistent with a broad anticipation pattern.

Detailed interpretation: Table 2 provides the scale and composition of the sample. Table 3 is the first direct test of the manipulation/anticipation story. If shorts were systematically creating or anticipating news, the short-volume ratio should rise before the news event. Instead, the event-time coefficients are small and do not show a systematic pre-news run-up.

How it fits the paper: These results move the paper away from private pre-news information and toward public-news processing. They also show why it is important to distinguish short volume from total trading volume: all trading volume spikes on news, but the short share does not spike before news.

Table 2

Summary statistics.

The database has 1,888,868 daily observations over the period January 3, 2005 through July 6, 2007 and it contains 3,167 NYSE-listed and traded securities including ordinary common shares in U.S. firms, ADRs, REITs, and closed-end funds. Panel A provides summary statistics at the firm level. News articles could be reissued throughout the day as more information becomes available; in such situations, we consider all of the related article updates to be one unique news event and we keep track of the number of articles that are rolled into this unique news event. *News articles per firm-day* is a count of all news articles including reissued (updated) articles, while *Unique news events per firm-day* is a count of the unique stories, excluding subsequent updates to an article. *Short vol./Total vol.* is short volume from the NYSE TAQ Regulation SHO database as a fraction of total volume; *Exempt* and *Non-exempt* denote market maker short sales (exempt) from non-market maker short sales (non-exempt); see Section 3.2.1 for details. Market capitalization is from CRSP. Panel B contains summary statistics on the frequency of each news category in the database; news articles can be classified into more than one category at the same time.

Panel A: Firm level statistics	Mean	Median	1st Percentile	99th Percentile	Standard deviation
News articles per firm-day	1.264	0.000	0.000	20.000	4.331
Unique news events per firm-day	0.027	0.000	0.000	13.000	2.934
Short vol./Total vol.	0.196	0.175	0.005	0.624	0.272
Short vol./Total vol.—Exempt	0.036	0.014	< 0.001	0.323	0.081
Short vol./Total vol.—Non-exempt	0.176	0.157	0.004	0.553	0.268
Market capitalization (\$ mm)	\$5,857	\$1,228	\$33	\$80,368	\$19,331
Panel B: News categories	N		News categories		N
10K	1,183		Initial public offerings		9,166
8K	10,202		Insider stock buys		20,215
Acquisitions, mergers, takeovers	55,994		Insider stock sells		51,199
Analysts' comments & ratings	48,743		Joint ventures		9,071
Annual meetings	3,871		Labor issues		12,302
Antitrust news	5,091		Lawsuits		14,784
Bankruptcy-related filings	5,979		Leveraged buyouts		2,171
Bond ratings & comments	14,925		Management issues		13,321
Buybacks	5,971		Market news		14,259
Contracts, Defense	4,406		Money market news		1,209
Contracts, Government (not defense)	3,294		New products & services		24,434
Contracts, Nongovernment	19,050		Personnel appointments		29,602
Corporate governance	4,725		Point of view		15,487
Corporate restructurings	5,519		Product distribution		2,456
Divestitures or asset sales	11,265		Research & development		5,541
Dividend news	23,653		Spinoffs		1,734
Earnings	41,556		Stock options		5,777
Earnings projections	36,774		Stock ownership		25,462
Financing agreements	6,816		Stock splits		1,895
High-yield issuers	168,803				

(a) Table 2: Summary statistics. The table documents sample size, short-volume ratios, and news category frequencies.

Table 3

Regression analysis of short-volume ratio around news events.

The table contains the results of panel data regressions that examine short sales volume around news events for NYSE-listed and traded securities over the period January 3, 2005 through July 6, 2007. In each regression the dependent variable is daily firm short volume divided by total firm volume and the independent variable of interest is an indicator variable that takes the value one if a news story occurs and zero otherwise, regardless of whether there is news on any of the surrounding days. In Panel A we examine short sales volume around all news events, while in Panels B and C we examine short sales volume around negative and positive news events, respectively. We define a news event as negative (positive) if the announcement day return is in the bottom (top) quintile of returns on a given day, respectively. In each panel we examine six different regressions that vary the timing of the dependent variable relative to the news event to examine short-volume changes around news. For example, $t-2$ indicates that the dependent variable is observed two days prior to the news event. *After minus before* indicates that the dependent variable is the difference in short volume after the event relative to before the event. All regressions include fixed effects by firm and month. To control for the documented response of short sellers to past returns, we include two lags of daily returns (measured in decimals, where the lags are relative to the timing of the dependent variable). *** indicates significance at the 1% level, ** indicates significance at the 5% level, and * indicates significance at the 10% level.

	Event time of the dependent variable					After minus before
	$t-2$	$t-1$	$t=0$	$t+1$	$t+2$	
Panel A: All news						
Return $_{t-2-\text{day lag}}$	0.388***	0.385***	0.388***	0.389***	0.390***	0.606***
Return $_{t-1-\text{day lag}}$	0.271***	0.273***	0.274***	0.274***	0.274***	-0.453***
News event	-0.001**	-0.002***	< 0.001	0.001	< 0.001	0.003***
N	1,490,754	1,496,281	1,522,378	1,522,378	1,522,378	1,439,881
Adj. R ²	0.161	0.163	0.161	0.161	0.161	0.009
Panel B: Negative news						
Return $_{t-2-\text{day lag}}$	0.388***	0.385***	0.340***	0.392***	0.390***	0.605***
Return $_{t-1-\text{day lag}}$	0.271***	0.273***	0.274***	0.274***	0.277***	-0.452***
News event	0.001	0.003***	-0.019***	0.002**	0.002*	-0.002
N	1,490,754	1,496,281	1,522,378	1,522,378	1,522,378	1,439,881
Adj. R ²	0.161	0.163	0.161	0.161	0.161	0.009
Panel C: Positive news						
Return $_{t-2-\text{day lag}}$	0.388***	0.385***	0.388***	0.386***	0.390***	0.601***
Return $_{t-1-\text{day lag}}$	0.271***	0.273***	0.274***	0.274***	0.274***	-0.452***
News event	< 0.001	-0.004***	0.002***	0.001*	< 0.001	0.007***
N	1,490,754	1,496,281	1,522,378	1,522,378	1,522,378	1,439,881
Adj. R ²	0.161	0.163	0.161	0.161	0.161	0.009

(b) Table 3: Regression analysis of short-volume ratio around news events. The key evidence is the absence of a systematic pre-news rise in the short-volume ratio.

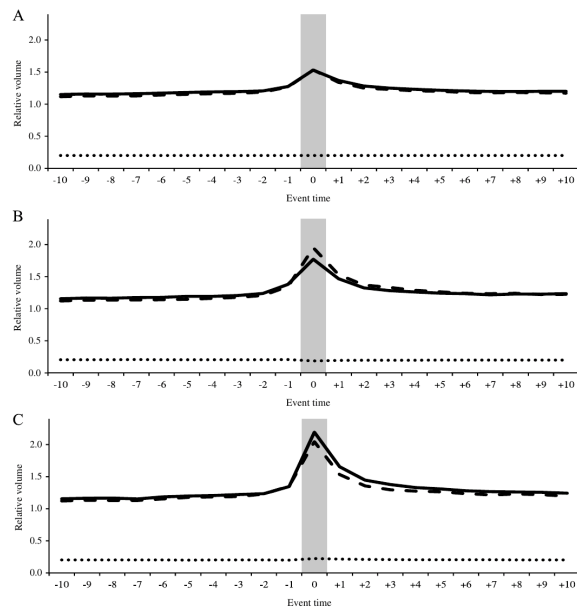
6.3 Figure 2: Daily volume around news events; Table 4: Cross-sectional relation between monthly percentage returns, short sales, and news

Read:

- Figure 2 shows that both short volume and total volume jump on the event day, but the short-volume ratio stays nearly flat.
- The ratio pattern holds for all news, negative news, and positive news.
- Table 4 shows the standard negative relation between short-volume ratio and future returns.
- The interaction of short volume with news is negative and significant: return predictability becomes stronger on news days.
- The interaction with negative news is especially large: about -1.805 for raw returns and -1.758 for adjusted returns.

Detailed interpretation: Figure 2 reinforces the timing evidence: news days are high-activity days, but short sellers do not dominate the pre-news run-up. Table 4 then provides the main profitability result. The short-volume ratio predicts lower future returns, and this predictability strengthens substantially when a news event occurs.

How it fits the paper: The combination of Figure 2 and Table 4 is the core logic of the paper: shorts are not generally early, but they are much more informative when public information is present.



(a) Figure 2: Daily volume around news events. Short volume and total volume spike together on news days, while the short-volume ratio is nearly flat.

Table 4

Cross-sectional relation between monthly percentage returns, short sales, and news.
The table contains Fama and MacBeth (1973) regression results examining the relation between returns, short sales, and news for NYSE-listed and traded securities over the period January 3, 2005 through July 6, 2007. For each model, we run 609 daily cross-sectional regressions and we take the time-series mean of the coefficients and use the standard deviation to estimate standard errors. The dependent variable is the buy and hold (compound) return percent over the subsequent 20 trading days. For models 1-3 the dependent variable is raw returns, while models 4-6 use size- and momentum-adjusted returns. Short-volume ratio is daily short volume/total volume and Size is the log of market capitalization lagged by one day. We define Negative, Neutral, and Positive news event variables that equal one if a news event occurs and the announcement day return is in the bottom quintile, middle three quintiles, or top quintile of returns on a given day, respectively. Short vol. $\times$ News is the product of Short-volume ratio and the News event indicator. Return_{it-1} is the return (in decimals) on each stock the day before short volume and news are observed. T-statistics are below the parameter estimates in italics and are calculated using Newey-West (1987) standard errors with 20 lags. *** indicates significance at the 1% level, ** indicates significance at the 5% level, and * indicates significance at the 10% level.

	Dependent variable: Raw returns (in percent)			Dependent variable: Adjusted returns (in percent)		
	[1]	[2]	[3]	[4]	[5]	[6]
Intercept	1.712*** (4.70)	1.719*** (4.75)	1.721*** (4.94)	0.541*** (5.32)	0.546*** (5.02)	0.548*** (5.21)
Short-volume ratio	-0.529** (-2.19)	-0.453* (-1.90)	-0.509** (-2.20)	-0.686*** (-3.27)	-0.597*** (-2.82)	-0.652*** (-3.19)
News event		-0.020 (-0.23)			0.004 (0.05)	
Short vol. $\times$ News		-0.479** (-2.16)			-0.561** (-2.51)	
Size			-0.006*** (-2.76)			
Negative news event			0.206 (1.37)			0.134 (1.05)
Neutral news event			0.021 (0.25)			-0.095 (-1.31)
Positive news event			0.384** (2.41)			0.329** (2.32)
Short vol. $\times$ Neg. news			-1.805*** (-4.26)			-1.758*** (-4.14)
Short vol. $\times$ Neut. news			-0.597*** (-2.72)			-0.327 (-1.35)
Short vol. $\times$ Pos. news			-1.113** (-2.40)			-1.155*** (-2.74)
Return _{it-1}			2.591 (1.62)			2.900* (1.86)
Return _{it-2}			3.566** (1.91)			3.393** (1.91)

(b) Table 4: Cross-sectional relation between monthly percentage returns, short sales, and news. The negative short-sale signal becomes much stronger on news days, especially negative news days.

6.4 Figure 3: Cumulative portfolio returns; Table 5: Monthly percentage returns from portfolios conditioned on short selling and news

Read:

- Figure 3 translates the regression evidence into cumulative trading-strategy performance.
- Short selling alone generates a meaningful long-short return, but adding news improves performance.
- Conditioning on positive/negative news produces the steepest cumulative return path.
- Table 5 reports monthly portfolio returns: short sales alone have a mean long-short return of 1.43%; short sales plus news have 2.02%; short sales plus news sign have 5.94%.

Detailed interpretation: The portfolio exercise shows economic magnitude. A strategy based on short selling alone performs well, but combining short selling with the presence and sign of news increases profitability sharply. The key distinction is not merely news versus no news; it is whether short sellers are on the correct side of news that the market subsequently processes slowly.

Economic message: The trading strategy results make the mechanism tangible. If public news created equal information for everyone, conditioning on news should not dramatically improve short-sale profitability.

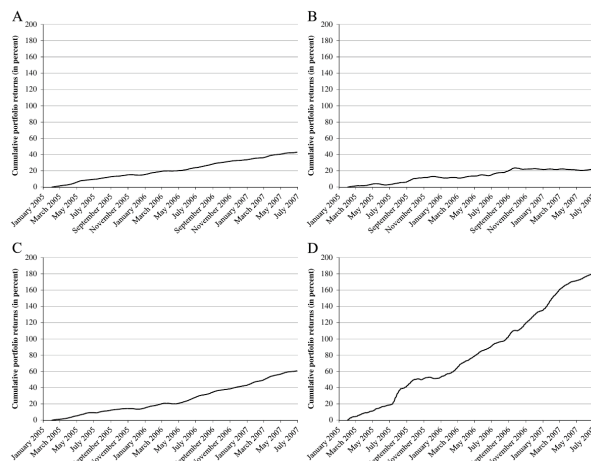


Fig. 3. Cumulative portfolio returns for NYSE-listed and traded securities over the period January 3, 2005–July 6, 2007. The figure displays cumulative returns from long-short portfolios formed by conditioning on short sales and/or news events. Each day, portfolios are formed by conditioning on the previous day's level of short sales and/or by conditioning on the occurrence of news events the previous day. These equal-weighted portfolios are then held for 20 days and the portfolio formation process is repeated each day so that 1/20 of the portfolio is effectively rebalanced each day, as in Boehmer, Jones, and Zhang (2008). The total daily portfolio return is thus the mean return calculated across the 20 separate positions that are open at each point in time. Summary statistics are calculated across the individual daily returns and are multiplied by 20 to approximate a monthly return and all statistics are shown in percent. In Panel A, the conditioning variable is the previous day's level of short sales, such that firms with short sales in the lowest (highest) quintile are assigned to the long (short) portfolio. In Panel B, firms with positive (negative) news are assigned to the long (short) portfolio. In Panel C, firms that experienced news and had short sales in the lowest (highest) quintile are assigned to the long (short) portfolio. In Panel D, firms with positive (negative) news and short sales in the lowest (highest) quintile are assigned to the long (short) portfolio.

(a) Figure 3: Cumulative portfolio returns. The strongest cumulative gains come from jointly conditioning on short selling and positive/negative news.

Table 5

Monthly percentage returns from portfolios formed by conditioning on short selling and/or news events.

The table contains mean monthly percentage returns for portfolios calculated over the period January 3, 2005 through July 6, 2007. Each day, portfolios are formed by conditioning on the previous day's level of short sales and/or by conditioning on the occurrence of news events the previous day. These equal-weighted portfolios are then held for 20 days and the portfolio formation process is repeated each day, so that 1/20 of the portfolio is effectively rebalanced each day, as in Boehmer, Jones, and Zhang (2008). The total daily portfolio return is thus the mean return calculated across the 20 separate positions that are open at each point in time. Summary statistics are calculated across the individual daily returns and are multiplied by 20 to approximate a monthly return and all statistics are shown in percent. In Panel A, the conditioning variable is the previous day's level of short sales, such that firms with short sales in the lowest (highest) quintile are assigned to the long (short) portfolio. In Panel B, firms with positive (negative) news are assigned to the long (short) portfolio. In Panel C, firms that experienced news and had short sales in the lowest (highest) quintile are assigned to the long (short) portfolio. In Panel D, firms with positive (negative) news and short sales in the lowest (highest) quintile are assigned to the long (short) portfolio.

	Long	Short	Long-short
Panel A: Conditioning variable=Short sales only			
1st Percentile	-3.88	-6.33	-0.76
Median	2.75	1.53	1.45
Mean	2.40	0.97	1.43
99th Percentile	6.48	5.48	3.60
Panel B: Conditioning variable=News only			
1st Percentile	-7.84	-7.10	-2.89
Median	2.66	1.64	0.67
Mean	1.94	1.20	0.74
99th Percentile	7.41	8.28	5.33
Panel C: Conditioning variable=Short sales and news			
1st Percentile	-4.55	-8.25	-1.40
Median	3.01	1.11	2.04
Mean	2.64	0.62	2.02
99th Percentile	7.55	5.78	5.03
Panel D: Conditioning variable=Short sales and positive/negative news			
1st Percentile	-9.79	-12.93	-3.53
Median	5.98	0.80	5.30
Mean	6.24	0.30	5.94
99th Percentile	25.45	9.01	20.45

(b) Table 5: Monthly percentage returns from portfolios. The long-short return rises from 1.43% with short sales only to 5.94% when short sales and news sign are used together.

6.5 Table 6: Non-exempt trades; Table 7: Exempt trades

Read:

- Table 6 uses non-exempt short sales, which are more likely to be client trades.
- Non-exempt short-volume ratios strongly predict lower future returns.
- The non-exempt interaction with negative news is large and negative; this is the cleanest evidence of informed client short selling.
- Table 7 uses exempt trades, which are more likely to be market-maker trades.
- Exempt trades do not show the same news-processing pattern; in some specifications the interactions are positive.

Detailed interpretation: These two tables address the most important alternative explanation: perhaps short selling is mechanically generated by market makers who short to provide liquidity to optimistic buyers on news days. If that were true, exempt trades should drive the result. Instead, the result is concentrated in non-exempt trades. This shifts the interpretation toward informed client trading.

How it fits the paper: The trade-classification evidence links the return predictability to likely information-motivated traders rather than market-making mechanics.

Table 6

Cross-sectional relation between monthly percentage returns, short sales, and news for non-exempt trades.

The table contains Fama and MacBeth (1973) regression results examining the relation between returns, short sales, and news for NYSE-listed and traded securities over the period January 3, 2005 through July 6, 2007. The sample includes only those short sale transactions that were classified as non-exempt, as discussed in Section 3.2.1 of the text. For each model, we run 600 daily cross-sectional regressions and we take the time-series mean of the coefficients and use the standard deviation to estimate standard errors. The dependent variable is the buy and hold (compound) return percent over the subsequent 20 trading days. For models 1-3 the dependent variable is raw returns, while models 4-6 use size- and momentum-adjusted returns. Short-volume ratio is daily non-exempt short volume/total volume and Size is the log of market capitalization lagged by one day. We define Negative, Neutral, and Positive news event variables that equal one if a news event occurs and the announcement day return is in the bottom quintile, middle three quintiles, or top quintile of returns on a given day, respectively. Short vol. $\times$ News is the product of Short-volume ratio and the News event indicator. Return_{*t-1*} is the return (in decimals) on each stock the day before short volume and news are observed. *t*-statistics are below the parameter estimates in italics and are calculated using Newey-West (1987) standard errors with 20 lags. ***Indicates significance at the 1% level, **Indicates significance at the 5% level, and *Indicates significance at the 10% level.

	Dependent variable: Raw returns (in percent)			Dependent variable: Adjusted returns (in percent)		
	[1]	[2]	[3]	[4]	[5]	[6]
Intercept	2.013*** (5.57)	2.055*** (5.76)	2.090*** (6.15)	0.821*** (7.39)	0.858*** (7.28)	0.863*** (7.54)
Short-volume ratio	-0.869*** (-2.93)	-0.743** (-2.54)	-0.799*** (-2.79)	-1.025*** (-3.88)	-0.889*** (-3.40)	-0.976*** (-3.73)
News event		-0.192* (-1.96)			-0.161* (-1.84)	
Short vol. $\times$ News		-0.700*** (-2.69)			-0.741*** (-2.82)	
Size			-0.008*** (-3.84)			
Negative news event			0.070 (0.41)		-0.015 (-0.10)	
Neutral news event			-0.116 (-1.26)		-0.261*** (-3.22)	
Positive news event			0.292 (1.64)		0.208 (1.35)	
Short vol. $\times$ Neg. news			-2.578*** (-4.42)		-2.445*** (-3.95)	
Short vol. $\times$ Neut. news			-0.828*** (-3.15)		-0.437 (-1.63)	
Short vol. $\times$ Pos. news			-1.432*** (-2.88)		-1.408*** (-3.16)	
Return _{<i>t-1</i>}			5.082*** (2.20)		5.291*** (3.42)	
Return _{<i>t-2</i>}			6.029*** (4.06)		5.777*** (4.03)	

(a) Table 6: Cross-sectional relation for non-exempt trades. Non-exempt short selling predicts lower future returns and is especially informative around news.

Table 7

Cross-sectional relation between monthly percentage returns, short sales, and news for exempt trades.

The table contains Fama and MacBeth (1973) regression results examining the relation between returns, short sales, and news for NYSE-listed and traded securities over the period January 3, 2005 through July 6, 2007. The sample includes those short sale transactions that were classified as exempt, as discussed in Section 3.2.1 of the text. For each model, we run 600 daily cross-sectional regressions and we take the time-series mean of the coefficients and use the standard deviation to estimate standard errors. The dependent variable is the buy and hold (compound) return percent over the subsequent 20 trading days. For models 1-3 the dependent variable is raw returns, while models 4-6 use size- and momentum-adjusted returns. Short-volume ratio is daily exempt short volume/total volume and Size is the log of market capitalization lagged by one day. We define Negative, Neutral, and Positive news event variables that equal one if a news event occurs and the announcement day return is in the bottom quintile, middle three quintiles, or top quintile of returns on a given day, respectively. Short vol. $\times$ News is the product of Short-volume ratio and the News event indicator. Return_{*t-1*} is the return (in decimals) on each stock the day before short volume and news are observed. *t*-statistics are below the parameter estimates in italics and are calculated using Newey-West (1987) standard errors with 20 lags. ***Indicates significance at the 1% level, **Indicates significance at the 5% level, and *Indicates significance at the 10% level.

	Dependent variable: Raw returns (in percent)			Dependent variable: Adjusted returns (in percent)		
	[1]	[2]	[3]	[4]	[5]	[6]
Intercept	1.224*** (3.05)	1.254*** (3.13)	1.203*** (3.10)	0.318** (2.40)	0.313** (2.12)	0.306** (2.14)
Short-volume ratio	-3.492*** (-5.07)	-4.011*** (-5.88)	-3.924*** (-5.93)	-4.081*** (-6.22)	-4.360*** (-6.77)	-4.321*** (-6.70)
News event			-0.100 (-0.97)		0.019 (0.27)	
Short vol. $\times$ News			4.953*** (4.77)		3.023*** (2.72)	
Size			-0.003 (-1.51)			
Negative news event			-0.173 (-1.00)		-0.082 (-0.60)	
Neutral news event			-0.112 (-0.97)		-0.044 (-0.40)	
Positive news event			0.195 (1.64)		0.206* (1.90)	
Short vol. $\times$ Neg. news			8.288** (2.35)		3.892 (1.08)	
Short vol. $\times$ Neut. news			4.595*** (3.73)		2.944** (2.34)	
Short vol. $\times$ Pos. news			4.264 (1.52)		1.257 (0.42)	
Return _{<i>t-1</i>}			4.460* (1.78)		4.031* (1.80)	
Return _{<i>t-2</i>}			5.831** (2.50)		4.239** (2.06)	

(b) Table 7: Cross-sectional relation for exempt trades. Exempt trades do not deliver the same news-based information pattern, weakening the market-maker explanation.

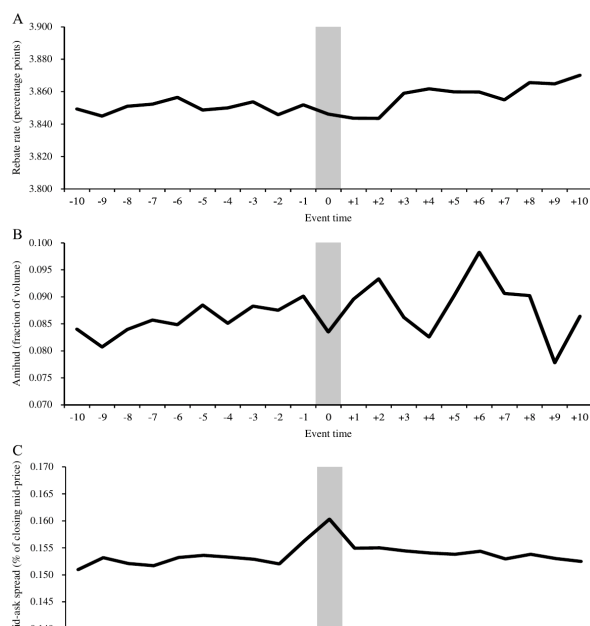
6.6 Figure 4: Market quality around news events; Table 8: Market quality around news events

Read:

- Figure 4 plots rebate rates, Amihud illiquidity, and bid-ask spreads around news events.
- Rebate rates do not improve on news days in a way that would make shorting easier.
- Amihud illiquidity does not fall on news days.
- Bid-ask spreads rise on the news day.
- Table 8 confirms these patterns using means and dependent *t*-tests.

Detailed interpretation: The liquidity hypothesis predicts cheaper shorting or lower transaction costs on news days. The data show the opposite for spreads and no meaningful improvement for the other market-quality measures. This is consistent with adverse-selection models: when informed trading is more likely, market makers widen spreads.

Economic message: The market quality evidence helps rule out an important non-informational channel. The stronger short-sale signal on news days is not simply because those days are cheaper days to short.



(a) Figure 4: Market quality around news events. Spreads rise on news days, while rebate rates and Amihud illiquidity do not show broad improvement.

Table 8

Market quality around news events.

The table displays the cross-sectional means of the *Rebate rate*, *Amihud illiquidity*, and *Bid-ask spread* measures in event time before, during, and after news events for NYSE-listed and traded securities over the period January 3, 2005 through July 6, 2007. $t-15$ represents the value 15 days before a news event, $t=0$ is the value on the day of a news event, and $t+15$ represents the value 15 days after a news event. Panel A contains the cross-sectional mean value and Panel B contains the p -values from a dependent t -test of differences in mean values relative to the news date ($t=0$). Because the subsamples around news dates do not all have the same number of observations, the differences presented in Panel B are not identically equal to the difference of the means presented in Panel A. The *Rebate rate* for an equity loan is the annualized rate (in percentage points) at which interest on collateral is rebated back to the borrower. The *Amihud (2002) illiquidity* measure is the daily illiquidity measure defined as $10^7 \times |ret_{it}|/volume_{it}$, where $volume_{it}$ is the dollar volume. *Bid-ask spread* is measured as a percentage of the closing mid-price on each day. ***Indicates significance at the 1% level, **indicates significance at the 5% level, and *indicates significance at the 10% level.

Event time	Rebate rate (in %)	Amihud illiquidity	Bid-ask spread
<i>Panel A: Mean</i>			
$t-15$	3.854	0.082	0.152
$t-10$	3.849	0.084	0.151
$t-5$	3.849	0.088	0.154
$t=0$	3.846	0.084	0.160
$t+5$	3.860	0.090	0.154
$t+10$	3.870	0.086	0.153
$t+15$	3.875	0.091	0.151
<i>Panel B: P-values from dependent t-test of differences</i>			
$t=0$ vs. $t-15$	0.042***	-0.001	0.003***
$t=0$ vs. $t-10$	0.029***	-0.001*	0.004***
$t=0$ vs. $t-5$	0.016***	< -0.001	0.002**
$t=0$ vs. $t+5$	-0.016***	< 0.001	0.006***
$t=0$ vs. $t+10$	-0.034***	< 0.001	0.005***
$t=0$ vs. $t+15$	-0.057***	< 0.001	0.007***

(b) Table 8: Market quality around news events. The table formalizes the event-time market-quality comparison around news days.

6.7 Table 9: Equity returns following specific news events

Read:

- Table 9 estimates the return-predictability regression separately for each of the 39 Dow Jones news categories.
- The short-volume coefficient is significantly negative for 13 categories.
- Categories with especially strong signals include earnings, earnings projections, high-yield issuers, new products and services, and stock ownership.
- Analysts' comments, bond ratings, divestitures/asset sales, and lawsuits also show strong negative coefficients.
- The Fisher test rejects the null that the category-specific coefficients are jointly uninformative.

Detailed interpretation: The category test shows that short sellers are not simply exploiting news as an undifferentiated trading-time marker. Their advantage is strongest in categories where interpretation, valuation judgment, or synthesis of public information is most valuable. The result also connects this paper to prior event-specific short-sale studies, but broadens the conclusion to many categories of public corporate news.

How it fits the paper: Table 9 completes the mechanism argument. It documents where information-processing skill appears to matter most and shows that the news-processing channel is not limited to earnings announcements.

Table 9

Equity returns following specific news events.

The table examines equity returns following news events for NYSE-listed and traded securities according to the model:

$$Ret_{it+1,t+20} = \alpha + \beta_1 \left(\frac{short\ vol_{t=0}}{market\ vol_{t=0}} \right) + \beta_2 Size_{t-1} + \beta_3 ret_{t-1} + \beta_4 ret_{t-2} + FE_i,$$

where the dependent variable is the buy and hold (compound) market-adjusted return percent from day 1 to day 20 following the news event and the short-volume ratio is short volume as a fraction of total volume on the day of the news event. Size is the log of market capitalization for each firm in billions, lagged by one day, and ret_{t-1} is the lagged return for each firm. The market-adjusted return is calculated net of the value-weighted market return from CRSP. Regressions are run individually for each news event and only when a news event occurs. As discussed in Section 3.2.2 of the text, to address the potential issue of news clustering, we remove stories in the same category that occur within 30 days of another story for the same firm. We include firm fixed effects in each regression. ***Indicates significance at the 1% level, **indicates significance at the 5% level, and *indicates significance at the 10% level.

News events	Short-volume ratio _{t=0}		Size _{t-1}		Return _{t-1}		Return _{t-2}		N
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	
10K	1.120	0.23	-0.216***	-3.20	-0.415	-1.16	1.589**	2.08	892
8K	-1.045	-0.79	-0.108***	-2.60	-0.081	-1.23	0.011	0.15	5,705
Acquisitions, mergers, takeovers	-1.568	-1.57	-0.336***	-5.92	0.035	0.50	-0.074	-1.01	8,643
Analysts' comments & ratings	-2.010**	-2.40	-0.392***	-8.17	-0.035	-0.84	0.025	0.47	11,032
Annual meetings	1.981	0.90	-0.037*	-1.79	0.087	0.78	0.153	0.88	2,382
Antitrust news	-0.675	-0.44	-0.079***	-4.47	-0.162	-0.94	-0.144	-0.89	2,027
Bankruptcy-related filings	2.023	0.59	-0.062*	-1.69	0.081	0.38	0.560***	4.60	1,513
Bond ratings & comments	-2.701**	-2.00	-0.122***	-4.46	-0.037	-0.79	0.283*	1.93	6,116
Buybacks	-1.310	-0.91	-0.065**	-2.36	-0.071	-0.71	0.039	0.38	3,323
Contracts, Defense	2.981	0.58	-0.068**	-2.53	-0.331	-0.95	0.249	1.20	589
Contracts, Government (not defense)	-2.620	-0.92	-0.130***	-2.58	-0.332	-0.65	0.170	0.74	1,224
Contracts, Nongovernment	0.115***	10.46	-0.210***	-4.77	-0.108	-1.00	-0.077	-0.80	5,547
Corporate governance	-2.424	-0.92	-0.079**	-2.48	0.057	0.32	0.139	1.35	1,862
Corporate restructurings	-0.729	-0.31	-0.066***	-3.16	0.317**	1.99	-0.020	-0.11	2,122
Divestitures or asset sales	-3.217**	-2.34	-0.082***	-3.50	-0.025	-0.26	0.007	0.05	3,905
Dividend news	-1.069*	-1.80	-0.138***	-4.23	0.030	0.61	-0.060	-1.36	13,358
Earnings	-2.762***	-4.25	-0.277***	-5.41	-0.065	-1.32	0.018	0.30	17,850
Earnings projections	-2.915***	-3.88	-0.340***	-6.60	0.001	0.02	-0.003	-0.05	13,127
Financing agreements	-3.371	-1.52	-0.165***	-4.60	0.057	0.32	-0.011	-0.03	3,353
High-yield issuers	-2.425***	-2.66	-0.556***	-5.58	0.052	1.00	0.126	1.47	9,649
Initial public offerings	1.375	0.79	-0.053**	-2.46	-0.014	-0.16	0.031	0.20	2,201
Insider stock buys	-1.741*	-1.83	-0.108**	-2.48	0.038	0.57	0.125*	1.92	7,511
Insider stock sells	-1.558*	-1.81	-0.145***	-5.18	0.045	0.49	-0.107	-0.91	9,250
Joint ventures	-0.441	-0.28	-0.159***	-4.51	-0.021	-0.18	-0.243*	-1.93	3,335
Labor issues	-0.243	-0.15	-0.103***	-3.12	0.128**	2.15	0.028	0.26	3,632
Lawsuits	-3.105**	-1.97	-0.223***	-5.38	0.194*	1.76	0.135	1.06	3,826
Leveraged buyouts	-4.019	-1.19	-0.024	-0.90	-0.113	-0.43	0.074	0.34	922
Management issues	-0.150	-0.11	-0.148***	-3.54	-0.061	-0.81	0.191	1.46	4,775
Market news	-0.138	-0.08	-0.224***	-4.79	-0.208	-0.93	0.183	1.57	3,644
Money market news	-1.850	-0.39	-0.083***	-4.07	-0.425**	-2.01	0.161	0.34	432
New products & services	-5.424***	-4.11	-0.153***	-4.36	0.091	0.56	-0.168*	-1.65	4,978
Personnel appointments	-0.982	-0.99	-0.261***	-4.52	-0.065	-0.97	0.178*	1.65	9,025
Point of view	0.070	0.04	-0.179***	-3.83	0.034	0.36	-0.313**	-2.53	3,721
Product distribution	-2.526	-1.02	-0.071**	-2.27	-0.246	-1.25	-0.195	-0.88	1,320
Research & development	-3.803	-1.08	-0.076**	-2.11	-0.084	-0.30	-0.104	-0.36	1,175
Spinoffs	-8.727	-1.34	-0.065	-1.63	-0.230	-0.87	-0.178	-0.47	738
Stock options	-1.420	-0.44	-0.116***	-5.90	-0.053	-0.47	0.167	0.98	2,544
Stock ownership	-1.805***	-2.83	-0.167***	-4.89	0.078	1.37	0.115**	2.19	16,677
Stock splits	-3.862*	-1.70	-0.044	-1.42	-0.009	-0.05	-0.017	-0.14	1,551
Fisher statistic	192.97***								
Fisher p-value	0.00%								

Table 9: Equity returns following specific news events. The table shows that short-sale predictability is concentrated in economically meaningful news categories and is jointly significant across categories.

7 Short summary

- Short sellers' information advantage is strongest around public news releases.
- They do not systematically increase relative shorting before public news, so the evidence does not support broad news anticipation or rumor creation.

- Short selling predicts lower future returns more strongly on news days and especially on negative news days.
- Portfolio strategies show large economic magnitudes when short selling is combined with the presence and sign of news.
- Non-exempt, likely client, short sales drive the findings; exempt, likely market-maker, trades do not.
- Market quality does not improve on news days; bid-ask spreads rise, which rejects a simple liquidity explanation.
- Category-specific tests show that the information-processing advantage is strongest for economically rich corporate news categories.
- The main message is that public information does not eliminate informed trading because traders differ in their ability to process public signals.

8 Conclusion

8.1 Core conclusion

The paper concludes that short sellers become informed in large part by processing public information better than other traders. The standard finding that short sales predict lower future returns is not just a generic short-sale effect. It is substantially amplified when public corporate news is released, and the amplification is strongest when the news is negative.

8.2 Economic intuition

If public information were immediately and equally understood by all traders, then short sellers would not earn especially high returns from trading on public news. The paper shows that this is not what happens. Public news increases the amount of information available, but it also increases the value of skill in interpretation. Short sellers appear to be among the traders who can convert noisy public disclosures into profitable trading signals.

8.3 Why the mechanism matters

The mechanism matters because it changes how we interpret both short selling and public disclosure. Short sellers are not merely holders of private information, nor are they primarily manipulators who create negative news. They are also information processors. This means that public disclosure can still produce informed trading if investors differ in attention, sophistication, and analytical skill.

8.4 Relation to market efficiency

The results are consistent with a semi-strong form of market efficiency that is imperfect in real time. News is public, but prices do not fully incorporate it immediately. The delayed response is not necessarily large enough to be costlessly arbitrated by all traders, but it is large enough that skilled short sellers can earn substantial gross returns around news events.

8.5 Limitations

The paper is not a randomized experiment and cannot observe traders' private research processes directly. Exempt status is an imperfect proxy for market makers, and portfolio returns are gross of full implementation costs. News classification and sentiment measures are noisy. However, the same interpretation is supported by

several independent tests: no pre-news shorting pattern, strong news-day return predictability, non-exempt trade concentration, rejection of the liquidity channel, and category-specific evidence.

8.6 Takeaway

The paper's lasting contribution is to show that public information can still be a source of informed trading. Short sellers' advantage is not simply that they know private facts earlier; it is that they process public news more effectively and trade on that interpretation before prices fully adjust.